National University of Computer and Emerging Sciences

Islamabad Campus

Probability and Statistics (MT2005)

BS(CS) and BS(CY)

Course Instructors

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Final-Exam

Total Time (Hrs): 3

Total Marks: 135

Total Questions: 7

Date: June 5, 2026

Roll No _____ Section _____ Student Signature _____

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Attempt all the questions

Note: All statistical tables are provided on page 9.

[CLO #1: Use concepts of Statistical methods for data analysis, frequency distribution, data presentation, measure of central tendency and variability, measure of dispersion, moments and skewness in computing domain.]

Question # 1 **[20 Marks]**

(a) The data show the monthly sales (in thousands of dollars) for seven months at two different stores.

Table 1: Monthly Sales

Location	1	2	3	4	5	6	7
Store X	610	228	1154	1159	2772	950	430
Store Y	275	1051	571	686	350	720	295

(i) Construct the boxplots for each dataset. (8)

(ii) Compare the IQR of the two distributions, and tell which one is more consistent. (1.5)

(iii) If the outlier is removed, would the range increase or decrease? Give reason. (1.5)

(b) The monthly electricity bills (in dollars) for 41 households in a city are summarized below.

Table 2: Frequency Distribution

Class	50–59	60–69	70–79	80–89	90–99	Total
Frequency (f)	5	12	14	6	4	41

(i) Construct the Histogram. (3)

(ii) Based on the shape of the histogram, comment on modality, symmetry, and skewness. (3)

(c) The average number of days that construction workers miss per year is 11. The standard deviation is 2.3. The average number of days that factory workers miss per year is 8 with a standard deviation of 1.8. Which class is more variable in terms of days missed? (3)

[CLO #2: Apply probability theory including sample space, joint probability, conditional probability, Bayes rule, total probability and independence on different software engineering problems.]

Question # 2 **[20 Marks]**

(a) Suppose that three types of antimissile defense systems are being tested. From the design point of view, each of these systems has an equally likely chance of detecting and destroying an incoming missile within a range of 250 miles with a speed ranging up to nine times the speed of sound. However, in actual practice it has been observed that the precision of these antimissile systems are not the same; that is, the first system will usually detect and destroy the target 10 out of 12 times, the second will detect and destroy it 9 out of 12 times, and the third will detect and destroy it 8 out of 12 times. We have observed that a target has been detected and destroyed. What is the probability that the antimissile defense system was of the third type? (5)

(b) A system is designed to detect whether a document is Original (O) or Plagiarized (P) using character trigrams. Features:

- X_1: High overlap of character trigrams (Yes / No)
- X_2: Unusual writing style shift (Yes / No)

Given:

$$\begin{aligned} P(P) = 0.3, \quad & P(O) = 0.7 \\ P(X_1 = \text{Yes} \mid P) = 0.8, \quad & P(X_1 = \text{Yes} \mid O) = 0.1 \\ P(X_2 = \text{Yes} \mid P) = 0.6, \quad & P(X_2 = \text{Yes} \mid O) = 0.2 \end{aligned}$$

(i) Compute the Naive Bayes score for plagiarism when both features are present. (2)

(ii) Classify the document. (2)

(iii) Explain how Naive Bayes can help detect plagiarism even if exact copying is avoided. (1)

(iv) Why is the classifier called Naive Bayes? (1)

(v) What happens if the independence assumption is violated? (1)

(c) Consider the following dataset for predicting whether a student will get a job offer or not. Here, the target variable is Y = Job Offer. Based on Information Gain for CGPA, Practical Knowledge, and Communication Skills, determine which attribute should be chosen as the root node of the decision tree. (8)

CGPA	Practical Knowledge	Communication Skills	Job Offer (Y)
> 3	Very Good	Good	Yes
> 2	Good	Moderate	Yes
> 3	Average	Poor	No
< 2	Average	Good	No
> 2	Good	Moderate	Yes
> 3	Good	Moderate	Yes
< 2	Good	Poor	No
> 3	Very Good	Good	Yes
> 2	Good	Good	Yes
> 2	Average	Good	Yes

[CLO #3: Transform given information to PMFs, PDFs and CDFs and express probability of events from statistical data.]

Question # 3 **[20 Marks]**

(a) Suppose the measurement error X of a certain physical quantity is described by the density function: (4)

$$f(x) = \begin{cases} k(3-x^2), & -1 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) Determine k that renders $f(x)$ a valid density function.

(ii) Find the probability that a random error in measurement is less than $\frac{1}{2}$.

(b) Let X be a random variable following the Gamma distribution with probability density function: (4)

$$f(x) = \begin{cases} \dfrac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Show that the expectation of X is given by $E(X) = \frac{\alpha}{\lambda}$.

(c) Let $X \sim \text{Exponential}(\lambda)$. Find the moment generating function (MGF) of the Exponential distribution. (4)

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \ge 0 \\ 0, & x < 0 \end{cases}$$

(d) Let X be a normally distributed random variable with mean $\mu = 50$ and standard deviation $\sigma = 10$. (4)

(i) Find the probability that X is less than 60.

(ii) Find the probability that $45 < X < 70$.

(e) An electronics company manufactures resistors that have a mean resistance of 100 ohms and a standard deviation of 10 ohms. The distribution of resistance is normal. Using the Central Limit Theorem, find the probability that a random sample of $n = 25$ resistors will have an average resistance of fewer than 95 ohms. (4)

[CLO #4: Apply the basic concepts of statistical inference and implement Z and t tests, regression and correlation analysis, and testing of regression coefficients using modern programming languages such as R or Python for statistical computing.]

Question # 4 **[20 Marks]**

(a) Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed (i.i.d.) random variables drawn from a population with mean μ and finite variance σ^2. Define the sample variance as:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2, \quad \text{where } \bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Show that sample variance is unbiased estimator of the population variance. (6)

(b) Suppose we have a random sample X_1, X_2, X_3 from a population with mean μ and variance σ^2. Two estimators of μ are proposed:

$$\hat{\mu}_1 = \frac{2X_1 + X_2 + 2X_3}{4}, \qquad \hat{\mu}_2 = \frac{2X_1 + 4X_2 + X_3}{4}.$$

(i) Derive the bias and variance of each estimator in terms of μ and σ^2. Then compute their Mean Squared Errors (MSE). (6)

(ii) Suppose the true population mean $\mu = 50$ and $\sigma^2 = 16$. Compute the numerical MSE for both estimators and state which estimator you would prefer if the goal is to minimize the MSE. Justify your answer. (3)

(c) Consider the probability density function:

$$f(x) = \frac{1}{\theta^2} x e^{-x/\theta}, \qquad 0 \leq x < \infty, \quad 0 < \theta < \infty$$

Find the maximum likelihood estimator for θ. (5)

[CLO #4: Apply the basic concepts of statistical inference and implement Z and t tests, regression and correlation analysis, and testing of regression coefficients using modern programming languages such as R or Python for statistical computing.]

Question # 5 **[14 Marks]**

(a) What is the correct frequentist interpretation of a 95% Confidence Interval? (2)

(b) A data scientist is comparing two variants of a neural network model for image classification:

- **Model A**: Standard architecture.
- **Model B**: Architecture with attention mechanism.

Both models are trained and evaluated on 30 independent test batches (each batch yields a single accuracy score). Summary statistics are as follows:

	Model A	Model B
Sample size	$n_A = 30$	$n_B = 30$
Sample mean	$\bar{x}_A = 84.2$	$\bar{x}_B = 87.5$
Sample variance	$s_A^2 = 12.5$	$s_B^2 = 9.8$

Assume the accuracies are approximately normally distributed. Use the given data to answer the following questions.

(i) Construct a 95% confidence interval for the true mean of Model A. Interpret the result. (3)

(ii) Construct a 99% confidence interval for the true mean of Model B. Interpret the result. (3)

(iii) Construct a 90% confidence interval for the difference in true means $(\mu_B - \mu_A)$. Assume the population variances are unknown but equal. State your assumption clearly. (6)

[CLO #4: Apply the basic concepts of statistical inference and implement Z and t tests, regression and correlation analysis, and testing of regression coefficients using modern programming languages such as R or Python for statistical computing.]

Question # 6 **[21 Marks]**

For each hypothesis test below, you must:

- State the null and alternative hypotheses.
- Compute the test statistic.
- Determine the critical value or p-value (as appropriate).
- State your conclusion in the context of the problem.

Use a significance level of $\alpha = 0.05$ for all tests.

(a) An AI researcher develops two reinforcement learning agents (Agent X and Agent Y) to play a complex strategy game. Each agent is run on 25 independent episodes, and the cumulative reward per episode is recorded. Summary statistics are:

	Agent X	Agent Y
Sample size	$n_X = 25$	$n_Y = 25$
Sample mean reward	$\bar{x}_X = 128.4$	$\bar{x}_Y = 135.7$
Sample variance	$s_X^2 = 156.25$	$s_Y^2 = 144.00$

Assume rewards are approximately normally distributed. Use the given data to answer the following questions.

(i) The researcher claims that the true mean reward of Agent X is less than 135. Test this claim. (3)

(ii) Test whether Agent Y has a significantly higher true mean reward than Agent X. Assume the population variances are equal. (3)

(b) A random sample of 100 medical school applicants at a university has a mean total score of 502 on the MCAT. According to a report, the mean total score for the school's applicants is more than 499. Assume the population standard deviation is 10.6. Is there enough evidence to support the report's claim at 1% level of significance using

(i) Critical Value Approach (3)

(ii) Confidence Interval Approach (3)

(c) A retail company wants to predict monthly sales revenue using two factors:

- X_1: Amount spent on online advertising (in thousands of dollars)
- X_2: Number of promotional emails sent to customers

The company collects data from $n = 30$ months and fits the following multiple linear regression model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

Coefficient Table

Parameter	Estimate	Std. Error
Intercept (β_0)	25	4
Advertising (β_1)	3.2	0.8
Emails (β_2)	1.5	0.6

It is also given that the values of $SSR = 1800$, $SSE = 900$, and $SST = 2700$.

(i) Test whether the amount spent on online advertising has a significant effect on monthly sales revenue. i.e. Test (β_1) (3)

(ii) Test whether the number of promotional emails sent has a significant effect on monthly sales revenue. i.e. Test (β_2) (3)

(iii) Calculate coefficient of determination and interpret it. (2)

(iv) Based on the value of R^2 and the significance of (β_1) and (β_2), would you recommend this model for prediction? (1)

[CLO #4: Apply the basic concepts of statistical inference and implement Z and t tests, regression and correlation analysis, and testing of regression coefficients using modern programming languages such as R or Python for statistical computing.]

Question # 7 **[20 Marks]**

(a) What is the fundamental difference between linear regression and logistic regression?(2)

(b) Why is the logistic function (sigmoid) appropriate for modeling probabilities? (2)

(c) Show that the derivative of the sigmoid function (6)

$$\sigma(z) = p = \frac{1}{1+e^{-z}} \text{ is given by} \qquad \frac{dp}{dz} = p(1-p)$$

(d) Prove that the inverse of the sigmoid function is the logit function, i.e., (4)

$$\sigma^{-1}(p) = z = \log\left(\frac{p}{1-p}\right).$$

(e) A marketing team wants to predict whether a customer will purchase a product (Yes = 1, No = 0) based on the amount spent on targeted online advertising (in thousands of Rupees). They collect data from $n = 50$ customers and fit a logistic regression model. The estimated coefficients are $\hat{\beta}_0 = -2.5$ and $\hat{\beta}_1 = 0.08$. Thus, the estimated log-odds are $\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = -2.5 + 0.08\,\text{AdSpend}$.

(i) What is the estimated probability of purchase for a customer with an ad spend of Rs. 50? (2)

(ii) Suppose the marketing team wants to increase the probability of purchase from 0.2 to 0.6. By how much should the ad spend increase? (4)

Good Luck!

PROBABILITY AND STATISTICS FORMULA SHEET

INFERENCE FOR SINGLE POPULATION

Test Statistics for Mean

Test Statistic $z = \dfrac{\bar{x}-\mu}{\sigma/\sqrt{n}}$ *and* $t = \dfrac{\bar{x}-\mu}{s/\sqrt{n}}$ *with* $d.f = n-1$

Critical Regions and P-Values for Z-test and T-test

	CR for $z-$*test*	*CR for* $t-$*test*	$P-$*values for* $z-$*test*	$P-$*values for* $t-$*test*
If H_1 *contains* $\neq$ *sign*	$\lvert z\rvert \geq z_{\frac{\alpha}{2}}$	$\lvert t\rvert \geq t_{\frac{\alpha}{2}(n-1)}$	$P = 2[1-\Phi(\lvert z_0\rvert)]$	$P = 2[\Phi(\lvert t_0\rvert)]$
If H_1 *contains* $<$ *sign*	$z < -z_\alpha$	$t < -t_{\alpha(n-1)}$	$P = \Phi(z_0)$	$P = 1-\Phi(\lvert t_0\rvert)$ *if* $t_0 > 0$ & $P = \Phi(\lvert t_0\rvert)$ *if* $t_0 < 0$
If H_1 *contains* $>$ *sign*	$z > z_\alpha$	$t > t_{\alpha(n-1)}$	$P = 1-\Phi(z_0)$	$P = \Phi(\lvert t_0\rvert)$ *if* $t_0 > 0$ & $P = 1-\Phi(\lvert t_0\rvert)$ *if* $t_0 < 0$

Confidence Intervals for Z-test and T-test

	Confidence Intervals for $z-$*test*	*Confidence Intervals for* $t-$*test*
If H_1 *contains* $\neq$ *sign*	$\bar{x} - Z_{\frac{\alpha}{2}}.\dfrac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\frac{\alpha}{2}}.\dfrac{\sigma}{\sqrt{n}}$	$\bar{x} - t_{\frac{\alpha}{2}(d.f)}.\dfrac{S}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}(d.f)}.\dfrac{S}{\sqrt{n}}$
If H_1 *contains* $<$ *sign*	$\mu \leq \bar{x} + Z_\alpha.\dfrac{\sigma}{\sqrt{n}}$	$\mu \leq \bar{x} + t_{\alpha(d.f)}.\dfrac{S}{\sqrt{n}}$
If H_1 *contains* $>$ *sign*	$\mu \geq \bar{x} - Z_\alpha.\dfrac{\sigma}{\sqrt{n}}$	$\mu \geq \bar{x} - t_{\alpha(d.f)}.\dfrac{S}{\sqrt{n}}$

INFERENCE FOR TWO POPULATIONS

Test Statistics for Difference of Means

Test Statistic $Z = \dfrac{(\bar{x_1}-\bar{x_2})-(\mu_1-\mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}}$ (*when* σ_1^2 *and* σ_2^2 *are known*)

$t = \dfrac{(\bar{x_1}-\bar{x_2})-(\mu_1-\mu_2)}{S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}}$ *with* $d.f = n_1+n_2-2$ *where* $S_p = \sqrt{\dfrac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}}$ (*when* σ_1^2 *and* σ_2^2 *are unknown*)

Critical Regions and P-Values

	CR for $z-$*test*	*CR for* $t-$*test*	$P-$*values for* $z-$*test*	$P-$*values for* $t-$*test*
If H_1 *contains* $\neq$ *sign*	$\lvert z\rvert \geq z_{\frac{\alpha}{2}}$	$\lvert t\rvert \geq t_{\frac{\alpha}{2}(n_1+n_2-2)}$	$P = 2[1-\Phi(\lvert z_0\rvert)]$	$P = 2[\Phi(\lvert t_0\rvert)]$
If H_1 *contains* $<$ *sign*	$z < -z_\alpha$	$t < -t_{\alpha(n_1+n_2-2)}$	$P = \Phi(z_0)$	$P = 1-\Phi(\lvert t_0\rvert)$ *if* $t_0 > 0$ & $P = \Phi(\lvert t_0\rvert)$ *if* $t_0 < 0$
If H_1 *contains* $>$ *sign*	$z > z_\alpha$	$t > t_{\alpha(n_1+n_2-2)}$	$P = 1-\Phi(z_0)$	$P = \Phi(\lvert t_0\rvert)$ *if* $t_0 > 0$ & $P = 1-\Phi(\lvert t_0\rvert)$ *if* $t_0 < 0$

Confidence Intervals for Difference of Means

	Confidence Intervals for $z-$*test*
If H_1 *contains* $\neq$ *sign*	$(\bar{x_1}-\bar{x_2}) - Z_{\frac{\alpha}{2}}.\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}} \leq \mu_1-\mu_2 \leq (\bar{x_1}-\bar{x_2}) + Z_{\frac{\alpha}{2}}.\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}$
If H_1 *contains* $<$ *sign*	$\mu_1-\mu_2 \leq (\bar{x_1}-\bar{x_2}) + Z_\alpha.\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}$
If H_1 *contains* $>$ *sign*	$\mu_1-\mu_2 \geq (\bar{x_1}-\bar{x_2}) - Z_\alpha.\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}$

	Confidence Intervals for $t-$*test*
If H_1 *contains* $\neq$ *sign*	$(\bar{x_1}-\bar{x_2}) - t_{\frac{\alpha}{2}(n_1+n_2-2)}S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}} \leq \mu_1-\mu_2 \leq (\bar{x_1}-\bar{x_2}) + t_{\frac{\alpha}{2}(n_1+n_2-2)}S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}$
If H_1 *contains* $<$ *sign*	$\mu_1-\mu_2 \leq (\bar{x_1}-\bar{x_2}) + t_{\alpha(n_1+n_2-2)}S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}$
If H_1 *contains* $>$ *sign*	$\mu_1-\mu_2 \geq (\bar{x_1}-\bar{x_2}) - t_{\alpha(n_1+n_2-2)}S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}$

SIMPLE LINEAR REGRESSION AND CORRELATION

$$r=\frac{n\left(\sum xy\right)-\left(\sum x\right)\left(\sum y\right)}{\sqrt{[n\sum x^2-\left(\sum x\right)^2][n\sum y^2-\left(\sum y\right)^2]}} \qquad For\ Testing\ Test\ Statistic\ is\ t=\frac{r\sqrt{n-2}}{\sqrt{1-r^2}}\ \ with\ d.f=n-2$$

$$For\ Y=\alpha+\beta X+\varepsilon,\quad \hat{\beta}=\frac{n\left(\sum xy\right)-\left(\sum x\right)\left(\sum y\right)}{n\left(\sum x^2\right)-\left(\sum x\right)^2}\ \ and\ \ \hat{\alpha}=\overline{Y}-\hat{\beta}\overline{X}\ \ also\ R^2=\frac{SSR}{SST}\ \ or\ \ R^2=1-\frac{SSE}{SST}$$

TESTING OF INTERCEPT α

$$Test\ Statistic\ t=\frac{\hat{\alpha}-\alpha}{S.E(\hat{\alpha})}\ \ where\ S.E(\hat{\alpha})=\sqrt{\frac{\sigma_e^{\,2}\sum X^2}{n.\sum\left(X-\overline{X}\right)^2}}\ \ also\ \sigma_e^{\,2}=\frac{\sum e^2}{n-2}\qquad CR:\left|t_{cal}\right|\geq t_{\frac{\alpha}{2}(n-2)}$$

$$The\ (1-\alpha)100\%\ confidence\ interval\ for\ \alpha\ is\qquad \hat{\alpha}-t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\alpha})\leq\alpha\leq\ \hat{\alpha}+t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\alpha})$$

TESTING OF INTERCEPT β

$$Test\ Statistic\ \ t=\frac{\hat{\beta}-\beta}{S.E\left(\hat{\beta}\right)}\ \ where\ \ S.E(\hat{\beta})=\sqrt{\frac{\sigma_e^{\,2}}{\sum\left(X-\overline{X}\right)^2}}\ \ also\ \sigma_e^{\,2}=\frac{\sum e^2}{n-2}\ \ and\ CR:\left|t_{cal}\right|\geq t_{\frac{\alpha}{2}(n-2)}$$

$$The\ (1-\alpha)100\%\ confidence\ interval\ for\ \beta\ is\qquad \hat{\beta}-t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\beta})\leq\beta\leq\ \hat{\beta}+t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\beta})$$

MULTIPLE LINEAR REGRESSION

$$For\ MLRM\quad Y=XB+E,\ we\ have\quad \hat{B}=(X^tX)^{-1}(X^tY),\ \hat{y}=X\hat{\beta},\ e=y-\hat{y}$$

$$SSE=e^te,\ SSR=\hat{B}^tX^tY-n\overline{Y}^2,\ SST=Y^tY-n\overline{Y}^2,\ R^2=\frac{SSR}{SST}\ \ or\ R^2=1-\frac{SSE}{SST}$$

$$Variance-Covariance\ Matrix\quad C=\sigma_e^2(X^tX)^{-1}\ \ where\ \sigma_e^2=\frac{SSE}{n-p}$$

$$For\ Testing\ of\ Parameters\ Test\ Statistic:\ \ t=\frac{\hat{\beta}_j-\beta_j}{se\left(\hat{\beta}_j\right)}\quad with\quad d.f=n-p$$

The $(1-\alpha)$ 100% **Confidence Interval** for β_j is given by

$$\hat{\beta}_j-t_{\frac{\alpha}{2}(n-p)}\times se\left(\hat{\beta}_j\right)\leq\ \beta_j\leq\ \hat{\beta}_j+t_{\frac{\alpha}{2}(n-p)}\times se\left(\hat{\beta}_j\right)$$

IMPORTANT PROBABILITY DISTRIBUTIONS

Probability Distribution	p.m.f or p.d.f	Mean	Variance
Uniform	$f(x)=1/b-a,\quad a\leq x\leq b$	$E(X)=(a+b)/2$	$V(X)=(a-b)^2/12$
Exponential	$f(x)=\lambda e^{-\lambda x},\quad x\geq 0$	$E(X)=1/\lambda$	$V(X)=1/\lambda^2$
Gamma	$f(x)=\frac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x},\quad x>0$	$E(X)=\alpha/\lambda$	$V(X)=\alpha/\lambda^2$
Beta	$f(x)=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1},\quad 0<x<1$	$E(X)=\alpha/\alpha+\beta$	$V(X)=\frac{\alpha.\beta}{(\alpha+\beta)^2.(\alpha+\beta+1)}$
Normal	$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2},\quad -\infty<x<\infty$	$E(X)=\mu$	$V(X)=\sigma^2$

Statistical Tables

Table 1: Customized Standard Normal Table for Selected Values

z	-0.5	-0.2	1.0	1.5	2.0	2.3	-2.5
$P(Z < z)$	0.3085	0.4207	0.8413	0.9332	0.9772	0.9900	0.0062

Table 2: Customized t-Table for Selected Degrees of Freedom

$t_\alpha(\text{df})$	$t_{0.025}(29)$	$t_{0.005}(29)$	$t_{0.05}(58)$	$t_{0.05}(24)$	$t_{0.05}(48)$	$t_{0.025}(27)$	$t_{0.05}(27)$
Critical Value	2.045	2.756	1.671	1.711	1.677	2.052	1.703

Table 3: Frequently Used Critical Values of Z

Significance Level (α)	0.10	0.05	0.01
z_α	1.282	1.645	2.326
$z_{\alpha/2}$	1.645	1.960	2.576